

ASSIGNMENT 1

Univariate and Multivariate Statistics

COURSE	FinTech 545 - Quantitative Risk Management
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Approach

Each problem is organized as Predict, Fit, and Reconcile. Predictions are based only on the exploratory view requested in the prompt; numerical model results are then reported separately and compared back to those predictions.

Headline findings: the Normal understates the left tail in Problem 1; Student-t errors dominate Normal errors in Problem 2; rank correlation correctly captures a curved monotone pair in Problem 3; the constant-width conditional band fails away from the center in Problem 4; and both ACF/PACF and AICc select AR(2) in Problem 5.

1. Reading the Shape of a Sample

Predict. Moments and candidate families

Before fitting a distribution, I used the sample mean, unbiased sample variance, bias-corrected skewness, and bias-corrected excess kurtosis. The sample is clearly left-skewed and heavy-tailed. Therefore a general normal inverse Gaussian (NIG) distribution is plausible: it can represent both negative skew and positive excess kurtosis. A Normal is ruled out by both the nonzero skewness and positive excess kurtosis. A symmetric Student-t can represent the positive excess kurtosis but is ruled out by the negative skewness. A lognormal is ruled out by the sign of skewness: every nondegenerate lognormal has positive skew. This judgment uses the requested moments alone; the observed negative values would independently rule out a lognormal.

Statistic	Estimate	Interpretation
Mean	0.00110439	Center is close to zero
Sample variance	0.00009625	Daily-scale dispersion
Skewness	-0.670	Negative: longer/heavier left side
Excess kurtosis	2.358	Positive: fatter tails than Normal

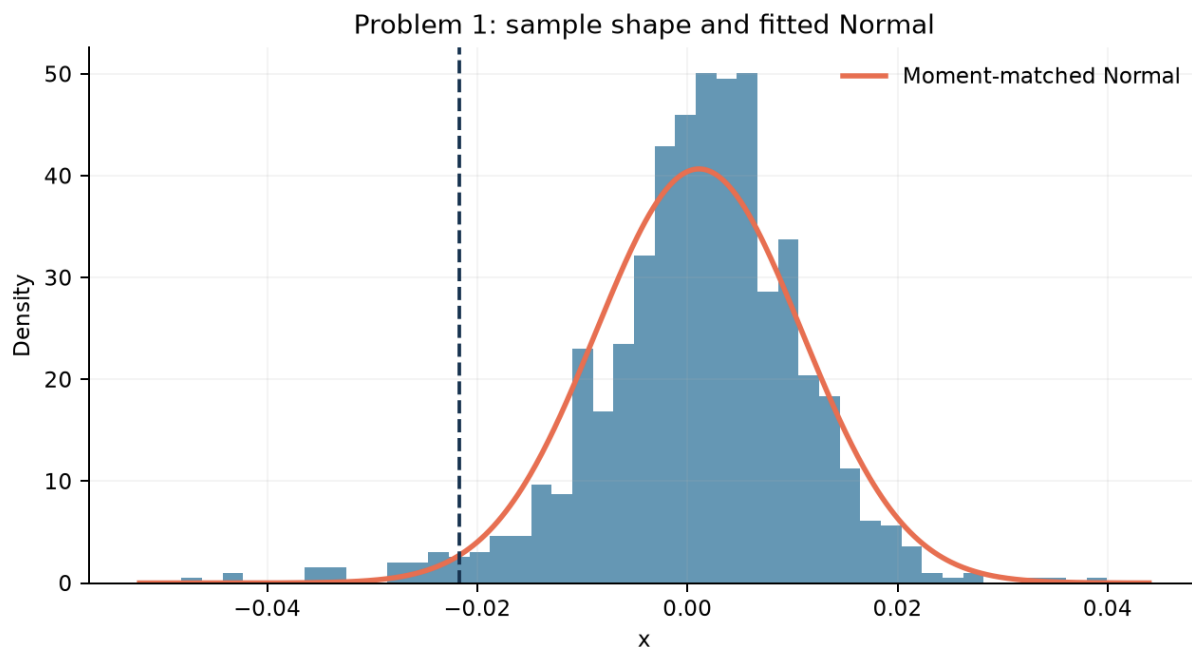


Figure 1. Histogram, moment-matched Normal density, and the fitted Normal 1% quantile (dashed).

Fit. Moment-matched Normal and its 1% quantile

Matching the sample mean and variance gives $N(0.001104, 0.00009625)$, with standard deviation 0.009811. Its 1% quantile is -0.021719. There are **26** observations below that threshold; a correctly specified 1% tail model should produce about **10** out of 1000 observations.

Reconcile. What the Normal gets wrong

The fit confirms the prediction. The Normal underestimates the probability of a large negative observation: the empirical count is 2.6 times the model expectation. Matching variance cannot repair the shape mismatch. The sample has both a heavier left tail and left asymmetry, while the Normal must allocate equal tail probability to both sides. Used as a risk model, it would understate downside tail risk.

2. A Regression Whose Errors Are Not Normal

Predict. Scatter and OLS assumption

The pre-fit scatter shows a strong positive, approximately linear center with no obvious curvature. Most points stay close to the line, but several have unusually large vertical deviations. I therefore expected a symmetric heavy-tailed error, specifically Student-t, rather than a Normal error.

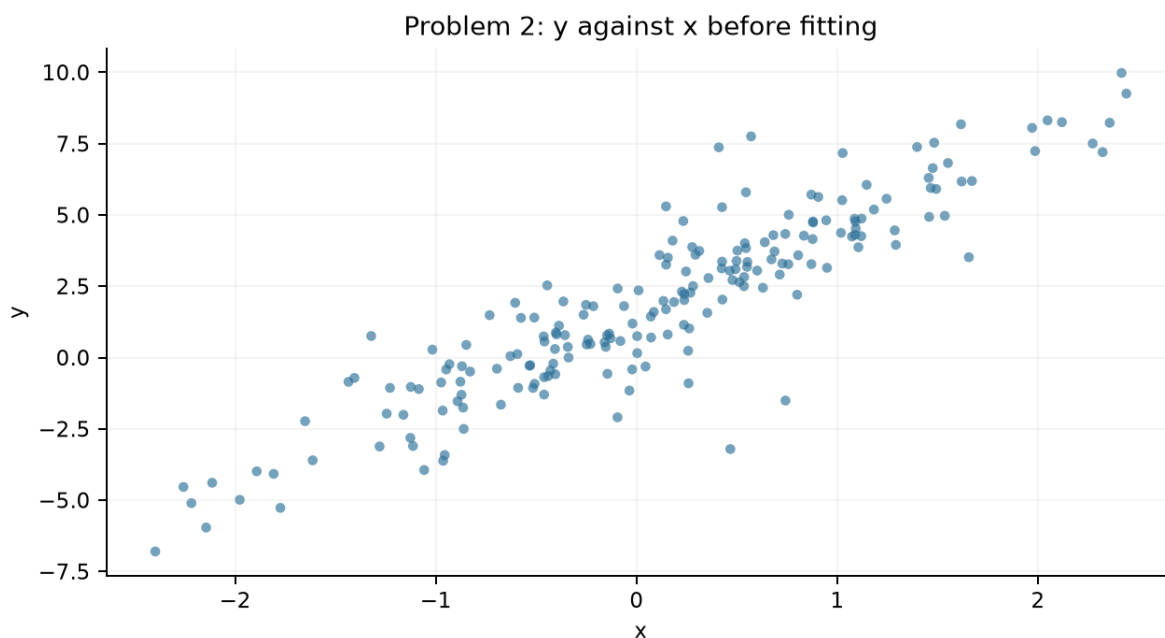


Figure 2. The required pre-fit scatter. The few large vertical deviations motivate a heavy-tailed error model.

This violates OLS assumption 7, normality of the error term. If assumptions 1-6 continue to hold, non-normality does not systematically bias the OLS slope: it remains unbiased/consistent. I therefore predicted little movement in beta. However, heavy tails make the slope less efficient and make exact finite-sample Normal/t inference invalid. Squared outliers can also make the usual OLS standard error noisy or inflated relative to a correctly specified heavy-tail likelihood.

Fit. OLS, Normal MLE, and Student-t MLE

Model	alpha	beta	Error parameters	AICc
OLS	1.5607	2.9908	s=1.3514	N/A
Normal MLE	1.5607	2.9908	sigma=1.3446	692.145
Student-t MLE	1.5480	3.0192	scale=0.9355; df=3.632	665.327

The OLS standard errors are $SE(\alpha)=0.0963$ and $SE(\beta)=0.0966$. OLS itself does not specify a likelihood, so AICc is not defined for bare OLS. Under Normal errors, the MLE line equals the OLS line; its scale differs because the MLE divides residual sum of squares by n. The Student-t AICc is lower by 26.817, decisive evidence in favor of the Student-t error.

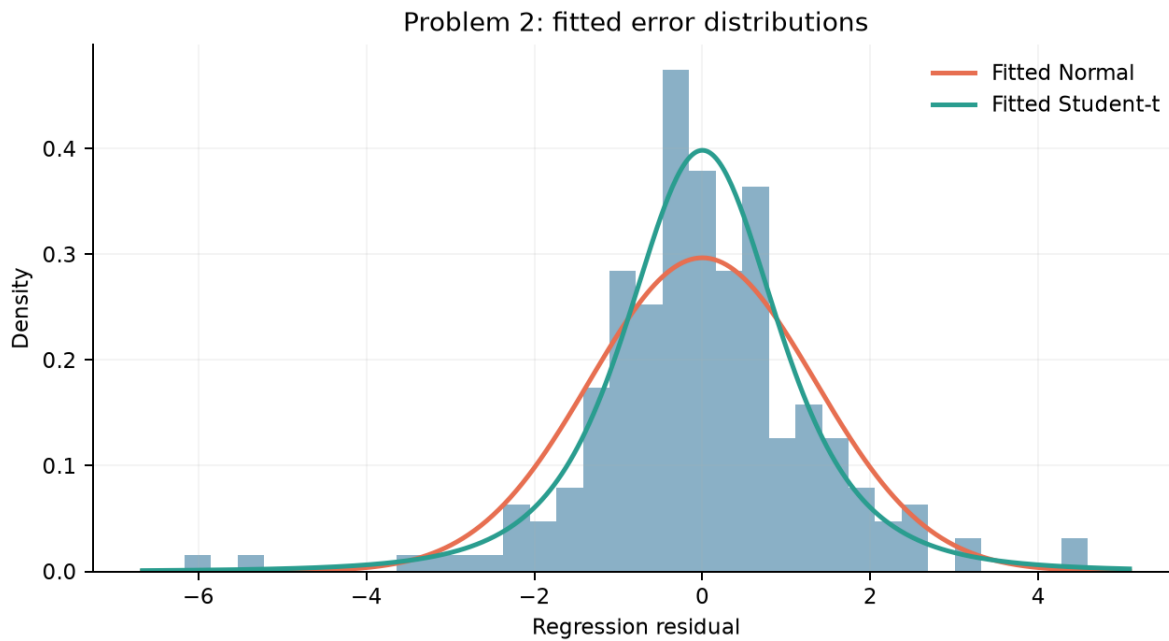


Figure 3. The Student-t fit is more peaked in the center and heavier in the far tails than the fitted Normal.

Reconcile. Slope stability, model failure, and capital quantiles

The beta estimates are 2.9908 (OLS and Normal MLE) and 3.0192 (Student-t MLE), a difference of only 0.0284. This agrees with my prediction: the violation primarily changes the error model and efficiency, not the central slope. The rejected Normal is getting the residual shape wrong. Its single scale is pulled outward by extremes, making it too broad in the shoulders and too flat at the center, yet it still assigns too little probability to very remote errors.

Upper error quantile	Normal	Student-t	Wider
95%	2.212	2.054	Normal
99.5%	3.464	4.620	Student-t

At 95%, the Normal is wider because its fitted sigma is inflated to compromise between the dense center and outliers. At 99.5%, the Student-t's polynomial tail dominates. For a high-confidence capital buffer, I would use the Student-t 99.5% quantile: it recognizes the extreme losses that the Normal understates. Because both fitted errors are symmetric, the corresponding lower-tail magnitudes are the same.

3. Pearson Against Spearman

Predict. Pairs expected to agree and disagree

Before calculating correlations, the pair plot suggested that x1 and x2 would produce the largest disagreement: their relationship is nearly deterministic and monotone, but strongly curved. I expected Pearson and Spearman to agree more closely for x1-x3 because that cloud is approximately linear, and for every pair containing x4 because those clouds show no relationship, linear or monotone. I expected some secondary disagreement for x2-x3 because the x2 transformation bends an otherwise monotone relationship.

Problem 3: pairwise relationships

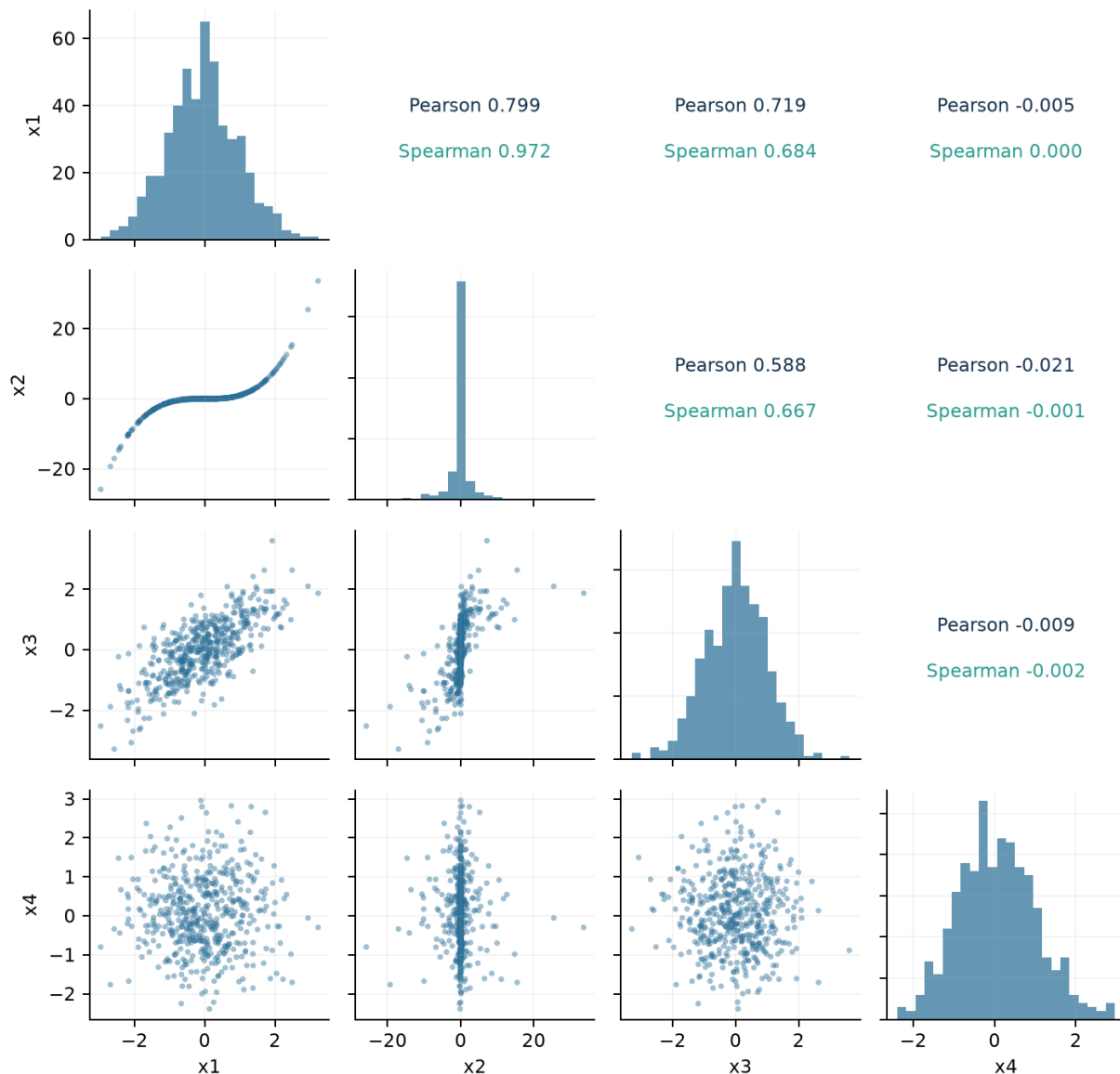


Figure 4. Every pair, with Pearson and Spearman coefficients displayed above the diagonal.

Fit. Correlation matrices

Pearson	x1	x2	x3	x4
x1	1.000	0.799	0.719	-0.005
x2	0.799	1.000	0.588	-0.021
x3	0.719	0.588	1.000	-0.009
x4	-0.005	-0.021	-0.009	1.000

Spearman	x1	x2	x3	x4
x1	1.000	0.972	0.684	0.000
x2	0.972	1.000	0.667	-0.001
x3	0.684	0.667	1.000	-0.002
x4	0.000	-0.001	-0.002	1.000

The largest absolute gap is for x1-x2: Pearson=0.799, Spearman=0.972, and the gap is 0.173.

Reconcile. Why the coefficients disagree

The plot shows x2 is approximately x1 cubed. Cubing preserves order, so nearly every pair of observations keeps the same rank and Spearman is close to one. The derivative of a cubic changes greatly across the range, so no single straight line represents the relationship equally well; Pearson is therefore lower. For the question 'how strong is the monotone association?', Spearman is the honest summary. Pearson is not wrong: it answers the narrower question 'how strong is the standardized linear covariation?' The observed results agree with the pre-fit prediction.

4. Conditional Distributions

Predict. Conditional variance and uncertainty reduction

I take the observed block to be $X_1=x_1$ and the target block to be $X_2=x_2$. Thus $\text{Sigma}_{11}=\text{Var}(x_1)$, $\text{Sigma}_{22}=\text{Var}(x_2)$, and $\text{Sigma}_{12}=\text{Sigma}_{21}=\text{Cov}(x_1,x_2)$. The sample covariance matrix is:

	x1	x2
x1	1.099875	1.696902
x2	1.696902	4.104749

The multivariate-Normal partitioned result is

$$\text{Var}(X_2 | X_1=a) = \text{Sigma}_{22} - \text{Sigma}_{21} \text{Sigma}_{11}^{-1} \text{Sigma}_{12}.$$

For this sample it is $4.104749 - (1.696902)^2/1.099875 = \mathbf{1.486746}$. The remaining variance factor is $1.486746/4.104749 = \mathbf{0.3622}$, so conditioning reduces variance by 63.78%. In standard-deviation units, uncertainty falls to $\text{sqrt}(0.3622) = 0.6018$ of its marginal level.

Under the multivariate Normal, this factor does not depend on the observed value a . The conditional variance formula contains only covariance blocks; a appears in the conditional mean, not in the conditional variance. I made this prediction before checking coverage against the data.

Fit. Conditional mean, band, and coverage

The corresponding conditional mean is

$$E[X_2 | X_1=a] = \mu_2 + \text{Sigma}_{21} \text{Sigma}_{11}^{-1}(a-\mu_1).$$

Numerically, $E[x_2|x_1=a] = 0.045288 + 1.542813(a - (-0.016714))$. The coefficient 1.542813 is exactly $\text{Cov}(x_2,x_1)/\text{Var}(x_1)$, the simple OLS slope from regressing x_2 on x_1 . The Normal 95% band uses conditional standard deviation 1.219322 and constant half-width $1.96s=2.389827$.

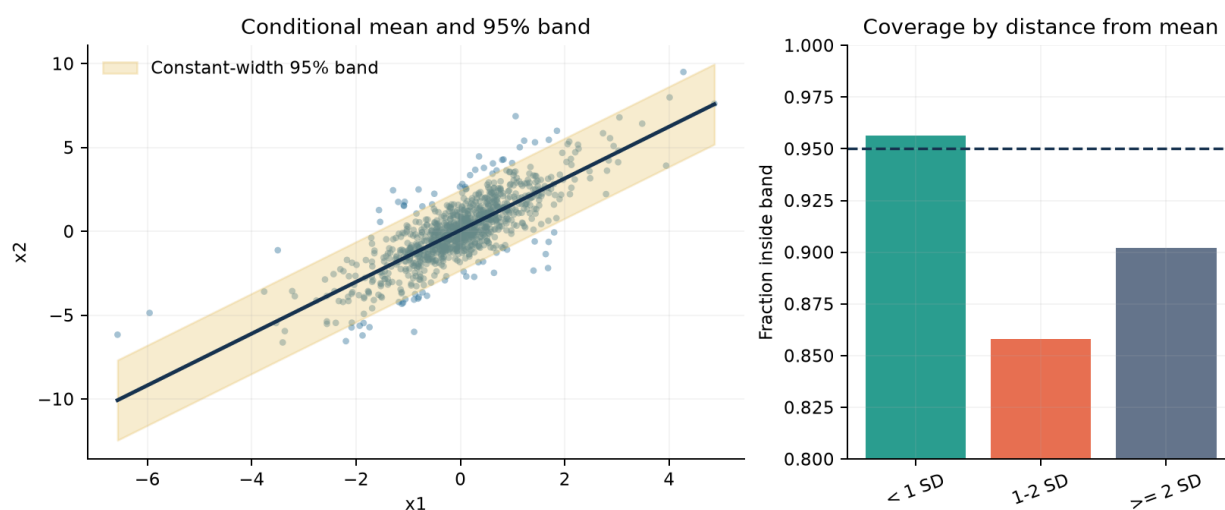


Figure 5. Conditional expectation, constant-width 95% band, and empirical coverage by distance bucket.

Bucket	n	Inside band	Residual SD
All observations	1000	93.50%	-
$ x_1 - \text{mean} < 1 \text{ SD}$	759	95.65%	1.076
$1 \leq x_1 - \text{mean} /\text{SD} < 2$	190	85.79%	1.580
$ x_1 - \text{mean} \geq 2 \text{ SD}$	51	90.20%	1.596

Reconcile. The failed constant-variance assumption

Overall, 935 of 1000 observations, or 93.50%, fall in the nominal 95% band. The aggregate is fairly close to 95%, but it hides a systematic pattern: coverage is 95.65% near the mean and only 85.79% and 90.20% in the two outer buckets. Residual standard deviation also rises from 1.076 near the center to about 1.59 away from it. This is evidence of heteroskedasticity: $\text{Var}(x_2|x_1)$ changes with x_1 , violating the constant conditional variance implied by a multivariate Normal.

What survives is the fitted central line as the OLS best linear projection, and it remains the conditional mean if $E[\text{error}|x_1]=0$ and the mean is truly linear. The slope interpretation as $\text{Cov}(x_2, x_1)/\text{Var}(x_1)$ also survives. What does not survive is the claim that covariance alone determines an exact Normal conditional distribution: the constant conditional variance, constant-width 95% band, and its uniform 95% coverage are invalid. Outside joint normality, the covariance formula is not automatically the exact conditional expectation; it is guaranteed only as the best linear predictor.

5. Identifying an AR or MA Order

Predict. ACF/PACF signature

The approximate 95% significance band is $\pm 1.96/\sqrt{500} = \pm 0.0877$. The ACF has a damped, oscillating pattern rather than a sharp cutoff. The PACF has two dominant spikes, lag 1 at 0.462 and lag 2 at -0.303, and then mostly cuts off. There are isolated later sample spikes, but not a persistent pattern; with many lags, some false 5% exceedances are expected. I therefore predicted an **AR(2)**. The rule used is: AR(p) has a decaying ACF and PACF cutoff after p; MA(q) reverses those roles.

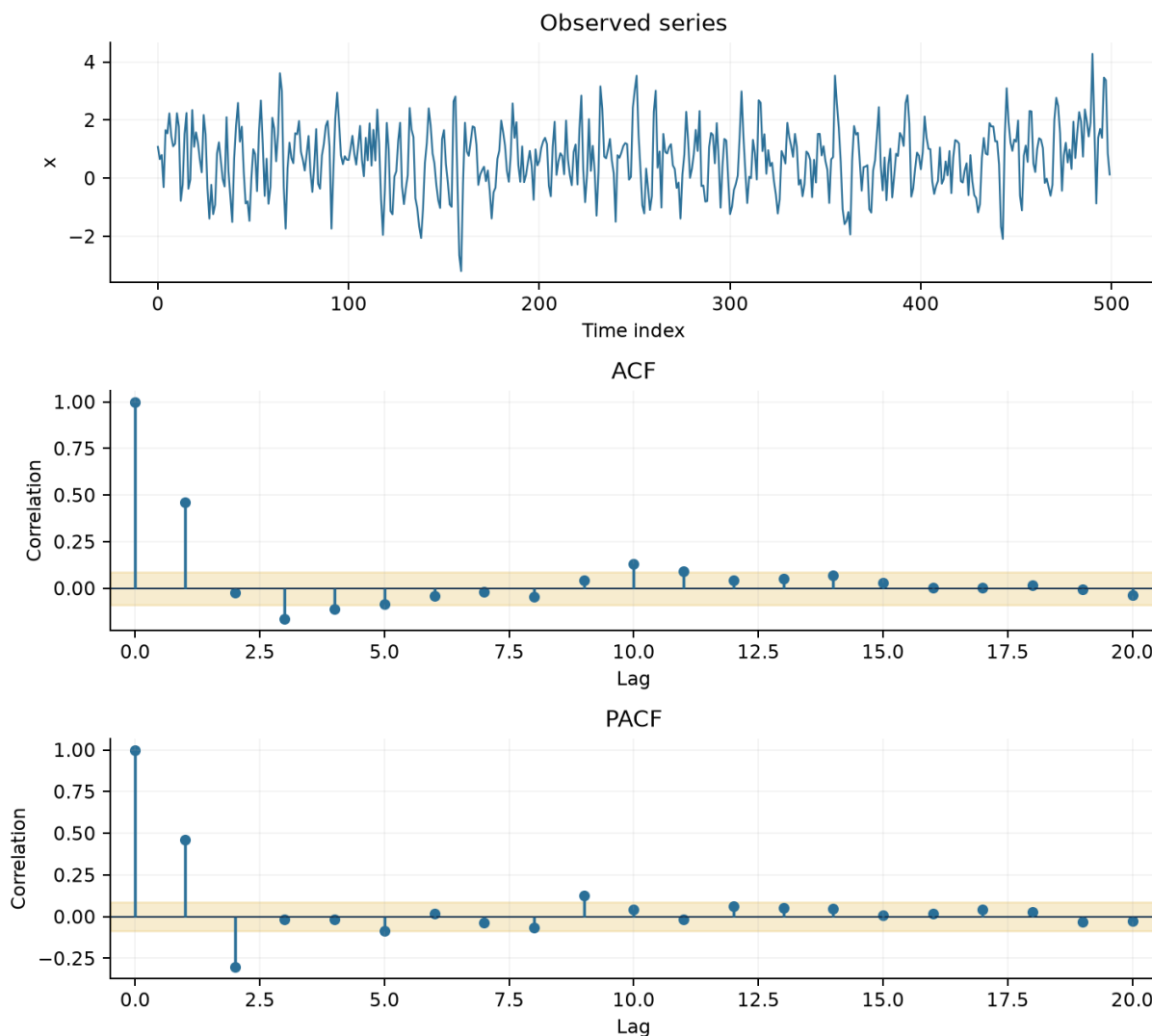


Figure 6. Series, ACF, and PACF. The shaded region is the $\pm 1.96/\sqrt{n}$ significance band.

Fit. AICc comparison

Model	AICc	Key coefficients
AR(1)	1418.587	ar.L1=0.4611
AR(2)	1372.438	ar.L1=0.6012, ar.L2=-0.3026
AR(3)	1374.343	ar.L1=0.5962, ar.L2=-0.2927, ar.L3=-0.0165
MA(1)	1389.084	ma.L1=0.5296
MA(2)	1381.232	ma.L1=0.6314, ma.L2=0.1615
MA(3)	1374.117	ma.L1=0.5931, ma.L2=0.0648, ma.L3=-0.1555

The smallest AICc is 1372.438 for AR(2), so the fitted model selection agrees with the pre-fit ACF/PACF prediction. Its coefficients are $\phi_1=0.6012$ and $\phi_2=-0.3026$.

Reconcile. AR(2) versus AR(3) and the role of complexity

AR(3) estimates $\phi_3=-0.0165$, which is very small. Adding it improves log likelihood only from -682.179 to -682.111, a gain of just 0.068. AICc rewards better likelihood but charges for every fitted parameter, with an extra finite-sample correction. The tiny likelihood gain cannot pay for the third AR coefficient, so AICc rises from 1372.438 to 1374.343. Plain R-squared has no parameter penalty and cannot decrease when an additional regressor is added. It therefore would not reject AR(3) on parsimony grounds, even though the added coefficient contributes almost no explanatory value.

Methods and Reproducibility

All calculations were produced by analysis.py from the five supplied CSV files. The script uses pandas/numpy for data handling, scipy for distributions and maximum likelihood, statsmodels for OLS and ARIMA estimation, and matplotlib for figures. AICc counts every fitted distribution/model parameter and uses $AICc = AIC + 2k(k+1)/(n-k-1)$, matching the Week 2 convention.

Quantity	Convention
Variance	Unbiased sample variance (ddof=1), except MLE scales
Skewness	Bias-corrected standardized sample skewness
Kurtosis	Bias-corrected excess kurtosis (Normal = 0)
Student-t	Location-scale t; df constrained above 2
PACF	Yule-Walker, no sample-size adjustment (statsmodels ywm)
AR/MA fits	Gaussian ARIMA likelihood with a constant and innovation variance

The README in this submission gives the exact commands required to install dependencies, reproduce results.json and every figure, and rebuild this PDF.